

Sensors and Image Formation

Examples

- A camera has a field of view of $90^\circ \times 90^\circ$ degrees, an image resolution of 400×400 pixels, and that the center of the image is the optical center of the camera. A point P has 3-D coordinates $(1\text{m}, 2\text{m}, 8\text{m})$ in camera coordinates. Find the pixel projection of point P in the image.

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- Solution: we first need to find f. With a fov of 90 degree, $f = \text{half the size of the image}$, or 200 pixels. Then use
 - $x_{\text{img}} = f X/Z + c_x = (200)(1\text{m}/8\text{m}) + 200 = 225$
 - $y_{\text{img}} = f Y/Z + c_y = (200)(2\text{m}/8\text{m}) + 200 = 250$

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 - Show that the width of the square in the image doesn't depend on the location
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- Solution
 - Let the plane be a distance Z from the camera
 - Let the sides of the square be at X_1 and X_2 ; $W = X_1 - X_2$
 - The image projection of those sides is $x_1 = f X_1 / Z$, $x_2 = f X_2 / Z$
 - The image width is $w = x_1 - x_2 = f W / Z$

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- Solution
 - Assuming the sensor elements are in a square grid, we have $\sqrt{10M}=3162$ elements on a side of 10 mm
 - So one sensor element is $10\text{mm}/3162 = 0.00316$ mm wide
 - At the center, angle is $\text{atan}(0.00316/6) = 5.27\text{e-}4$ radian

- What is the IFOV for the human eye? Assume one receptor cell on the retina is .003 mm wide, and the focal length is 17 mm
- What is width of smallest object you can see at 30m? Assume that the image of the object has to cover at least one receptor cell

- What is the IFOV for the human eye? Assume one receptor cell on the retina is .003 mm wide, and the focal length is 17 mm
 - Solution
 - $\text{atan}(.003/17) = 1.76\text{e-}4$ radian
- What is width of smallest object you can see at 30m? Assume that the image of the object has to cover at least one receptor cell
 - Solution
 - By similar triangles, $w/30\text{m} = .003/17 \rightarrow w = 0.0053 \text{ m}$